

PREDICTING OUTBREAK SEVERITY USING MACHINE LEARNING, DEEP LEARNING AND HYBRID MODELS

By

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ABSTRACT

Infectious disease outbreaks remain a major global challenge, and being able to predict the severity of such outbreaks has become even more important now. In this project, I explore how researchers have applied machine learning (ML), deep learning (DL), and hybrid models to forecast disease outbreak severity across different regions and diseases. I have reviewed 26 research papers that focused on diseases like COVID-19, dengue, influenza, and West Nile Virus, using models such as LSTM, CNN, CNN-LSTM hybrids, Graph Neural Networks, and Bayesian frameworks. Many of these studies also brought in extra data like climate patterns, people's movement, and even medical images to improve their predictions. From what I saw, hybrid models, especially CNN-LSTM and ensemble methods, work better than traditional models when it comes to accuracy. I also found that while these models offer higher accuracy, they come with challenges like computational demand and lack of interpretability. This review helped me identify key modeling strategies, limitations in current approaches, and future opportunities. I will use this information in the next stages of my Major Research Project, where I aim to design a severity prediction and risk mapping system using Toronto's outbreak data.

Keywords : Outbreaks, Severity Prediction, Machine Learning, Deep Learning, XGBoost, CNN-LSTM, Ensemble Models, Public Health, Infectious Diseases, Forecasting.

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1. INTRODUCTION

This document provides details on what domain, topic, dataset(s) and research question I chose for my Major Research Project (MRP). I start off by providing a brief background about the topic and datasets, then define the problem and state my research question, followed by literature review and a detailed exploratory analysis of my dataset. I then outline the methodology and experiments performed to train the classifiers, go over the results and close the paper with a discussion about this project and recommendations for future work.

1.1 Background

For my Major Research Project, I wanted to understand about prediction accuracy for disease outbreak severity using computational methods helping public health systems. COVID-19 outbreaks can show healthcare systems as overwhelmed, most especially when time is limited. Traditional methods often failed to capture real-time patterns, spatial spread, or the interactions of different risk factors. Therefore I used machine learning and Deep learning models.

In this literature review, I examined how researchers have used different modeling approaches of LSTM, CNN, BiLSTM, Random Forest, XGBoost, and hybrid models like CNN-LSTM, to forecast the severity of disease outbreaks. I looked at the ways that these models are used for various diseases. I also observed their methods for handling issues such as missing data, feature selection, model interpretability, and generalizability. Some studies improved prediction accuracy through integrated climate data, human mobility patterns, or remote sensing data.

This review intended to understand which techniques are effective as well as where research still has gaps. This knowledge is helpful to me in my own project, and I will then apply all these learnings as I develop a predictive model that can assess outbreak severity through using data directly from healthcare institutions located in Toronto.

2. LITERATURE REVIEW

2.1 RQ1: What are the key factors influencing outbreak severity?

The severity of infectious disease outbreaks is identified by a mix of variables. For instance, Santangelo et al. noted that traditional models often overlook variables such as climate change, and human mobility, which are crucial for forecasting severity accurately [7]. In contrast, newer models integrate high-resolution environmental data, satellite imagery, and even social media trends to improve forecasting accuracy [1][2].

A study by Liu et al. [1] points out that ML models, especially tree-based ones like Random Forest and XGBoost are capable of handling high-dimensional, correlated features that impact severity, such as population density, rainfall, and temperature. Similarly, the work by Gómez et al. [7] emphasizes the need for holistic models that combine data from clinical sources, climate models, and demographic records to truly capture disease dynamics and risk factors.

Moreover, research by Ong et al. [19] shows that vector indices and meteorological data such as rainfall and humidity are strong predictors in dengue outbreak models. Similarly, Rakovics et al. [13] showed that the best-performing models for COVID-19 severity prediction integrated chest X-ray data and age emphasizing that no single variable can capture outbreak severity in isolation.

2.2 RQ2: How do ML, DL, and hybrid models compare in terms of predictive accuracy?

Across the studies I reviewed, hybrid models consistently outperformed standalone ML or DL approaches in predicting outbreak severity. For instance, Zain and Alturki [11] used a CNN-LSTM hybrid that captured both spatial and temporal patterns of COVID-19 spread and achieved high accuracy. Similarly, Sebastianelli et al. [17] applied an ensemble of tree-based ML models and LSTM variants to forecast dengue, showing a significant improvement in early detection accuracy.

Machine learning models like XGBoost and SVM still perform well when working with structured tabular data. For example, Aleixo et al. [20] demonstrated the efficacy of explainable ML using SHAP values with XGBoost to predict dengue outbreaks in Brazil, offering both performance and transparency. However, these models often lack the temporal memory capabilities that deep learning models like LSTM bring.

On the deep learning side, LSTM and Bi-LSTM models have been widely used for temporal outbreak prediction. Alam et al. [12] used Bi-LSTM to forecast COVID-19 with superior performance compared to

traditional statistical models. Sembiring et al. [14] optimized LSTM parameters to improve accuracy and robustness across multiple disease datasets. But again, they tend to struggle with interpretability.

However, the best results came from hybrid systems. For example, Qaid et al. [10] combined deep learning models with tree-based ML classifiers for COVID-19 severity prediction, achieving robust performance. A similar hybrid was used by Park et al. [9], who developed a CNN-ML ensemble that integrated both radiological and clinical features. These systems blend the interpretability and efficiency of ML with the temporal and spatial memory of DL.

2.3 RQ3: Can feature engineering and advanced modeling techniques improve outbreak severity classification?

Absolutely, the studies show that targeted feature engineering and advanced model architectures significantly boost performance. For example, Chen and Moraga [21] integrated human mobility patterns with LSTM models, greatly enhancing their ability to predict dengue in Brazil. Similarly, Shah et al. [4] used large language models (LLMs) along with deep learning to extract semantically rich features from unstructured data such as clinical notes, improving predictive power.

Tasciotti et al. [25] proposed a Bayesian deep learning model to estimate COVID-19 superspreading events, and their results showed that including uncertainty estimates helped improve robustness and reduce overfitting. Meanwhile, Bonicelli et al. [26] used Graph Neural Networks (GNNs) trained on satellite imagery to predict West Nile Virus, showing that spatial modeling significantly enhances prediction.

Finally, Lucas et al. [15] demonstrated that even basic ML models can benefit from engineered spatiotemporal features. By incorporating lag variables, mobility indices, and weather conditions, they achieved better long-term outbreak predictions at the county level.

2.4 Conclusion of Literature Review

After reviewing, I understood in a much more deeper way how machine learning, deep learning, and hybrid models predict as well as classify the severity of infectious disease outbreaks. LSTM and CNN-LSTM deep learning models were in fact more effective in the capturing of complex temporal and spatial patterns while customary models such as Random Forest and XGBoost performed quite well on structured data. Hybrid models stood out since they were able to integrate a number of different data types and mostly provide improved accuracy. I also noticed many studies lacked transparency for how their models made decisions however few addressed real-world challenges like uncertainty estimation or

missing data. This highlights the need for predictive systems that are not just accurate, but also explainable and practical for use by healthcare teams.

My Major Research Project is supported in a direct way by perceptions like these. I now feel more confident to build a severity prediction system for Toronto's healthcare outbreak data using hybrid and interpretable models. My goal is to focus on models balancing performance with usability, and these models aid decisions in real-time public health settings.

3. DESCRIPTIVE ANALYTICS

As part of my Major Research Project, healthcare institutions throughout Toronto reported outbreak data that I began with. To understand reasons regarding various disease outbreaks such as respiratory, enteric, and others was the initial step. Since this analysis aids in understanding the patterns that are not visible, like what healthcare settings are most affected, how long outbreaks may last, and how outbreak characteristics can change over time, the analysis within this dataset is an important task. Certain perceptions are needed for data-driven decisions. These perceptions are critical for the process.

3.1 Dataset Overview

The dataset used was collected from the **City of Toronto Open Data Portal**, specifically from the Outbreaks in Toronto Healthcare Institutions. This dataset contains information about outbreaks in hospitals, long-term care homes (LTCHs), retirement homes, and other healthcare settings from **2016 to 2025**.

Key columns included:

- **Institution details** (name, address, setting type)
- **Outbreak information** (type, causative agents, start/end dates)
- **Status** (active/resolved) and derived **duration** in days
- Some duplicate columns were later resolved in preprocessing.

The dataset is available at Toronto Public Health Open Data ([Outbreaks in Toronto Healthcare Institutions - City of Toronto Open Data Portal](#))

3.2 Data Preprocessing

In this step, I cleaned and transformed the raw data. After that, I merged 10 CSV files into one complete dataset and added a "Year" column. While combining the files, I noticed that the columns like for causative agents had slightly different names in different years, so I fixed those and removed the duplicates. Since, each row did now have a proper unique ID, I did also add the id field.

I made a conversion of date columns to datetime format and then I made a calculation of the outbreak duration by subtracting the start date from the end date. I filled the missing values by -1, which keeps things clean. I cleaned text columns such as "Outbreak Setting," "Type of Outbreak," and "Active." I changed everything to lowercase and got rid of extra spaces to keep categories consistent.

My dataset had 6,101 rows at the end. It also would have had 12 helpful columns. I saved this final version as **preprocessed_dataset.csv**, and this is the dataset I used for all the EDA steps that followed.

3.3 Exploratory Data Analysis

The Exploratory Data Analysis (EDA) phase was important in order to understand the underlying patterns and characteristics of the outbreak data. This process helped me evaluate data quality, detect trends, identify outliers, and generate insights relevant for my predictive modeling goals.

3.3.1 Initial Structure

```

=== DataFrame Info ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6101 entries, 0 to 6100
Data columns (total 12 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   _id                                    6101 non-null   int64
1   Institution Name                      6101 non-null   object
2   Institution Address                   6101 non-null   object
3   Outbreak Setting                     6101 non-null   object
4   Type of Outbreak                    6101 non-null   object
5   Causative Agent-1                   6101 non-null   object
6   Causative Agent-2                   299 non-null    object
7   Date Outbreak Began                 6101 non-null   object
8   Date Declared Over                  6101 non-null   object
9   Active                              6101 non-null   object
10  Year                                6101 non-null   int64
11  Outbreak Duration (days)           6101 non-null   int64
dtypes: int64(3), object(9)
memory usage: 572.1+ KB
None

```

Figure 1 - Initial dataset structure

Figure 1 below shows the initial dataset structure, which contains 6,101 outbreak records and 12 columns. The columns capture details such as institution information, outbreak setting and type, causative agents, start and end dates, active status, year, and outbreak duration. Three columns are stored as integers, while the rest are stored as objects. Notably, **Causative Agent-2** has a high number of missing values compared to other fields, suggesting it is not always reported.

3.3.2 Descriptive Statistics and Distribution Checks

To better understand the nature of the outbreaks, I analyzed the Outbreak Duration column.

```

=== Numeric Summary for Outbreak Duration (days) ===
count      6101.000000
mean       17.014260
std        13.975621
min         1.000000
25%        10.000000
50%        14.000000
75%        19.000000
max        578.000000
Name: Outbreak Duration (days), dtype: float64

Skewness: 12.420997331415533
Kurtosis: 431.3215323400912

=== Value Counts for Outbreak Setting ===
Outbreak Setting
hospital-acute care      831
hospital-chronic care    717
hospital-psychiatric      98
ltch                     3391
retirement home         1035
shelter                   3
transitional care         26
Name: count, dtype: int64

=== Value Counts for Type of Outbreak ===
Type of Outbreak
enteric      512
other        21
respiratory  5568
Name: count, dtype: int64

=== Value Counts for Active ===
Active
n      6085
y       16
Name: count, dtype: int64

=== Value Counts for Year ===
Year
2016      268
2017      371
2018      439
2019      317
2020      553
2021      394
2022     1133
2023     1066
2024     1031
2025      529
Name: count, dtype: int64

```

Figure 2 - Summary Statistics and Value Counts for Key Variables

Figure 2 shows that most outbreaks lasted around 14 days, although a few extended beyond 500 days, as reflected in the skewness and kurtosis values indicating a right-skewed distribution.

In addition to the numeric field, I examined the distribution of several categorical variables. The majority of records came from long-term care homes (LTCHs), which was followed by retirement homes and hospitals. In terms of outbreak type, respiratory outbreaks were the most frequently reported. Most cases I found had been resolved when I checked the active status though a few cases stayed active. The data finally showed outbreak activity peaked within 2022 and 2023, likely because of the COVID-19 pandemic.

3.3.3 Univariate Visualizations

Univariate analysis focuses on the exploration of only one single variable. It helps me understand how individual features in the dataset like outbreak duration, setting, or type are distributed as well as patterned and also how frequently they occur, without consideration for their relationship with other variables. This step is important since it allows detection of skewness, identification of outliers, and observation of dominant trends. Such observations can affect feature engineering plus model selection later in the project.

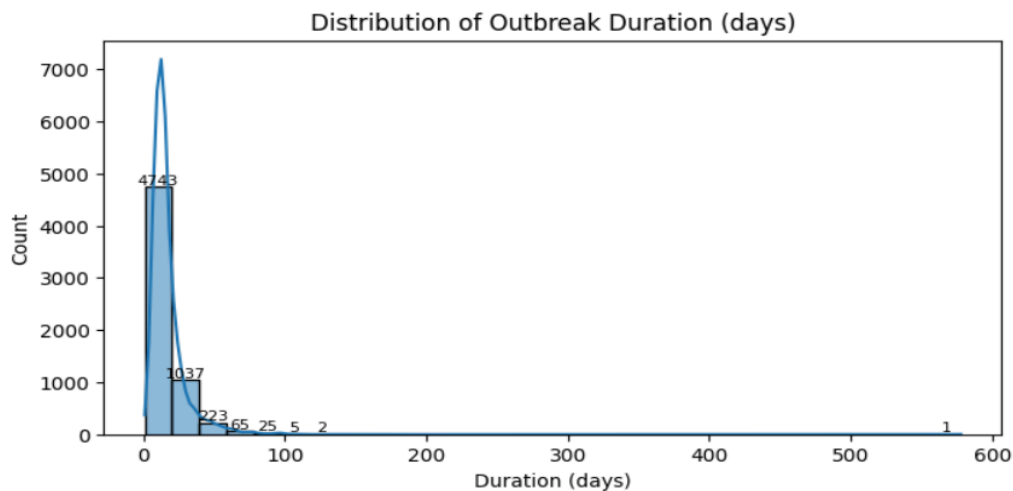


Figure 3 - Distribution of Outbreak Duration (Days)

Figure 3 shows a strong right skew in outbreak durations, with most outbreaks lasting fewer than 30 days and only a few extending beyond 100 days.

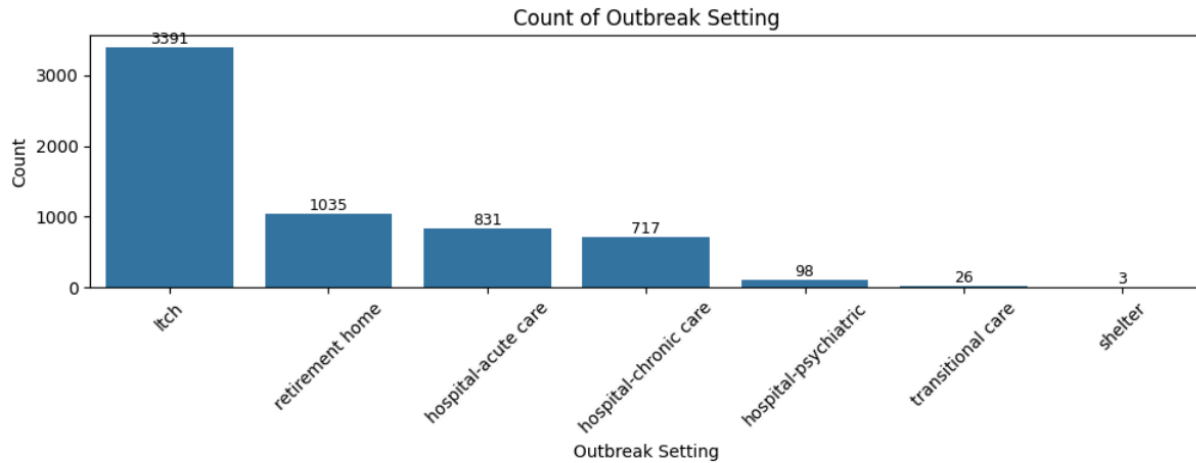


Figure 4 - Count of Outbreak setting

For the categorical features, bar plots helped illustrate the distribution clearly. I observed that long-term care homes (LTCHs) reported the highest number of outbreaks, followed by retirement homes and hospitals (Figure 4).

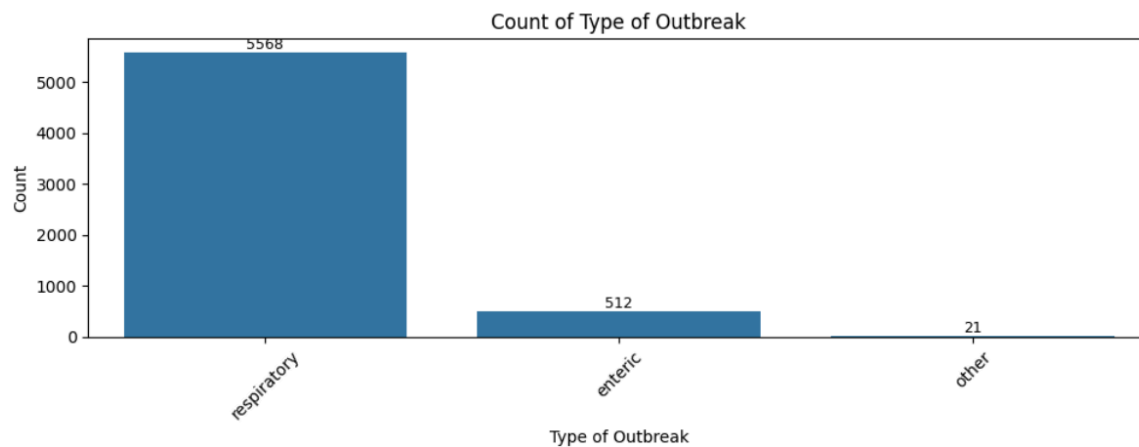


Figure 5 - Count of Type of Outbreak

In terms of outbreak type, respiratory outbreaks were the most common (Figure 5).

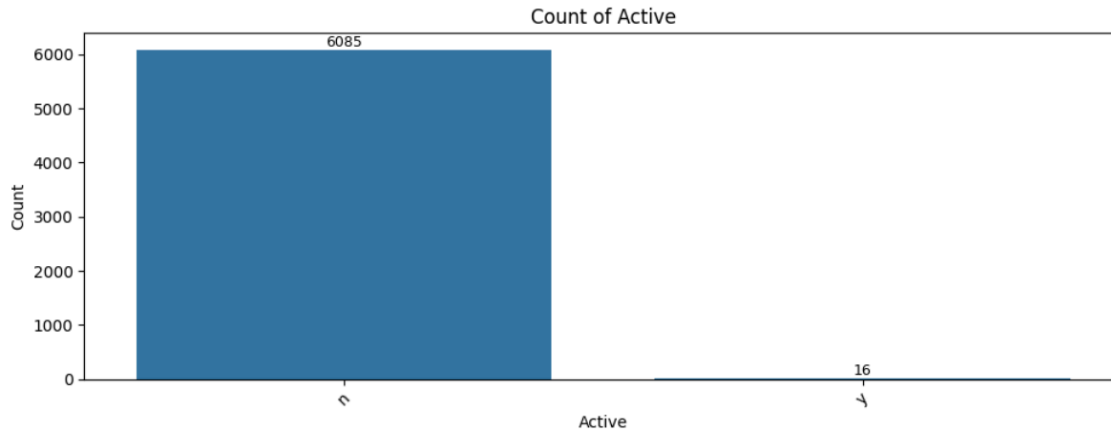


Figure 6 - Count of Outbreak Status

Additionally, Figure 6 shows that most outbreaks were marked as resolved (Active = n), with only a small fraction still ongoing.

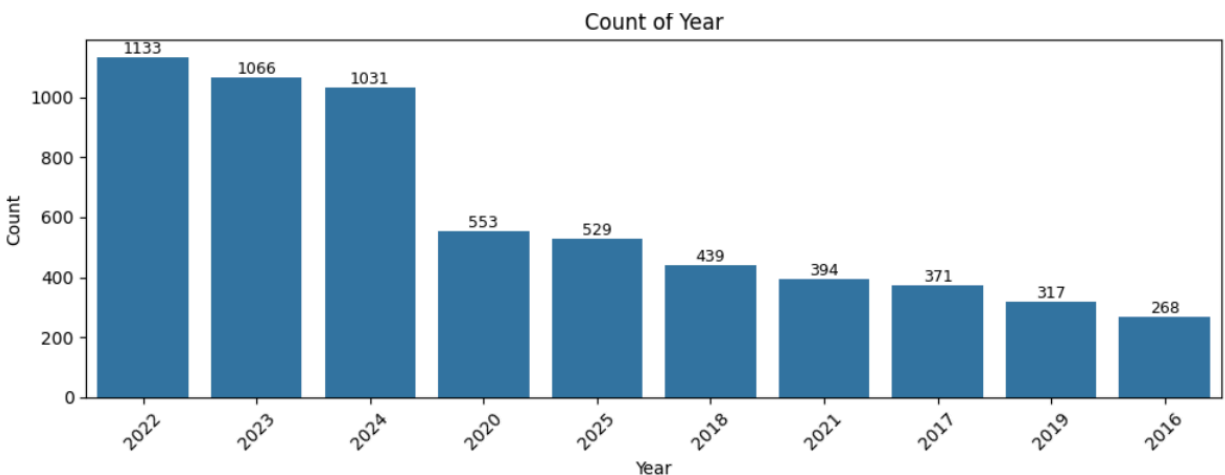


Figure 7 - Yearly distribution of outbreaks

Lastly, a year-wise breakdown showed that outbreak counts peaked between 2022 and 2024, likely reflecting the impacts of the COVID-19 pandemic during that period.(Figure 7)

3.3.4 Bivariate Visualizations

Bivariate analysis studies the relationship existing between two variables only. This step is important for offering an important understanding. It aids you in the revealing of patterns or of associations that are not visible when you are examining single variables. In my case, it did help me to understand just how outbreak duration is influenced by other features such as outbreak type and setting and year or if the

outbreak still is active. These perceptions help to select features especially also to decide which variables could predictively empower my model.

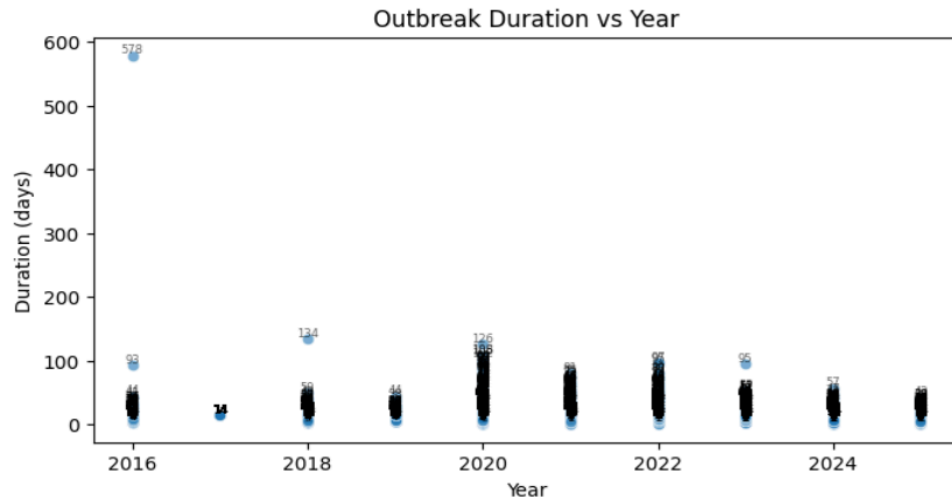


Figure 8 - Outbreak Duration vs Years

Figure 8 shows that most outbreaks stayed under 30 days across all years, but a few in 2020–2021 lasted much longer, reflecting COVID-19’s prolonged impact.

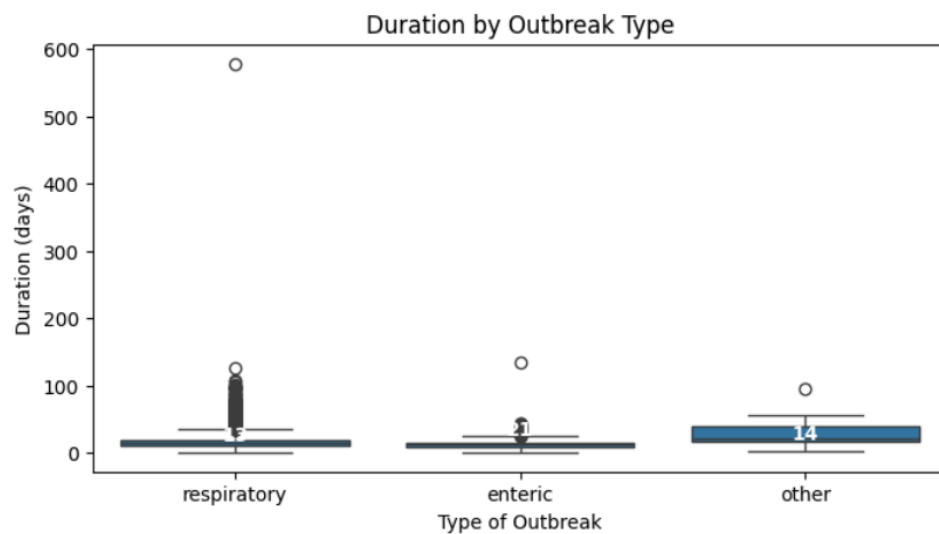


Figure 9 - Duration by Outbreak Type

Next, I compared duration by outbreak type which showed that respiratory and enteric outbreaks had similar median durations, but respiratory had more variation (Figure 9).

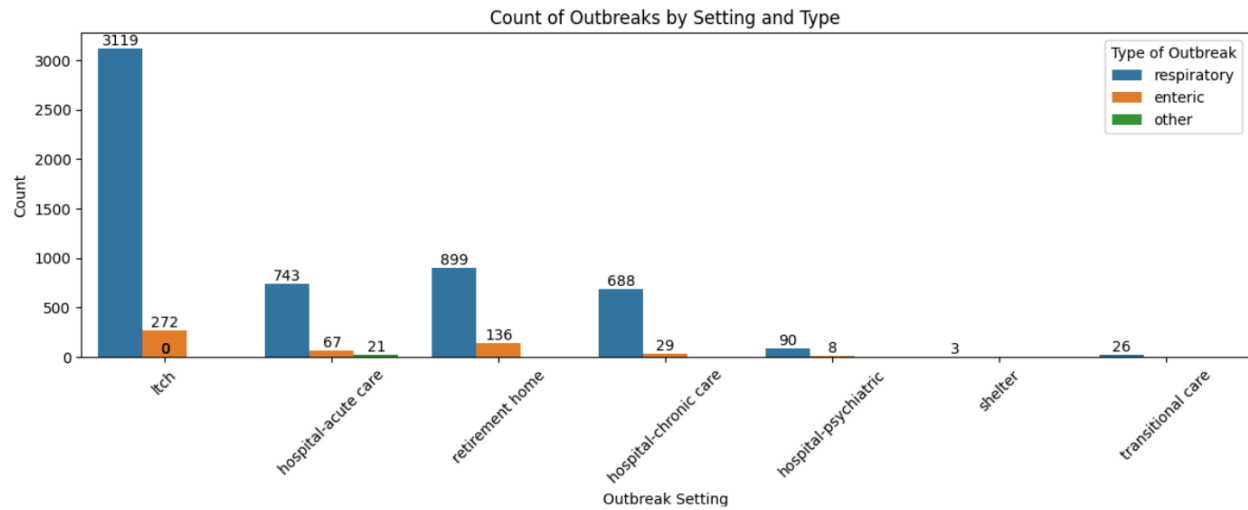


Figure 10 - Count of Outbreak by Setting and Type

Lastly, Figure 10 shows that. Long-term care homes had the highest number of respiratory outbreaks, followed by hospitals and retirement homes.

3.3.5 Correlation Analysis for Numeric Features

Correlation analysis helps me understand whether two numeric variables move together in a meaningful way. It's a quick check to see if one variable could potentially explain or predict another.

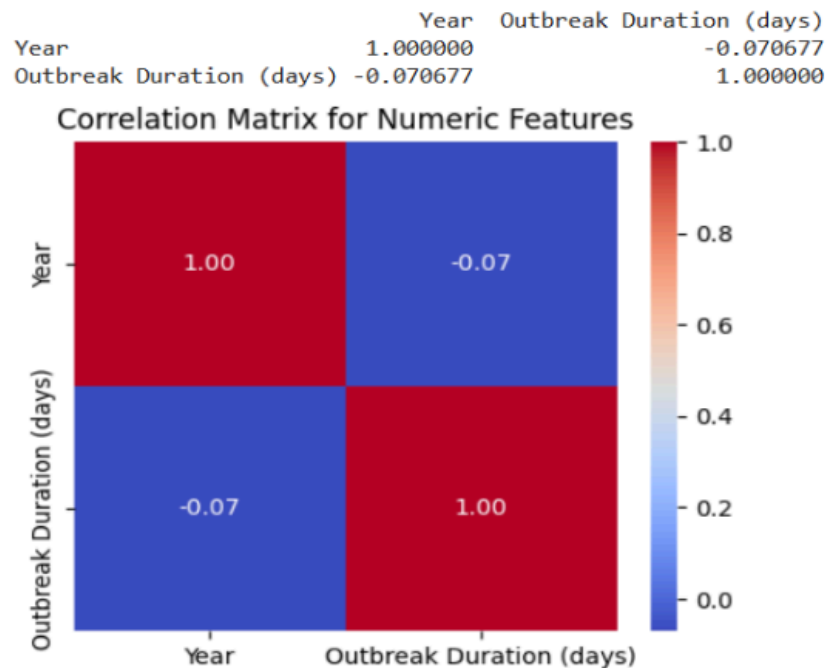


Figure 11 - Correlation Matrix for Numeric Features

The heatmap in Figure 11 correlates the year for outbreak with the duration for outbreak in days. The correlation coefficient was calculated at about -0.07. This is extremely close to zero. This means that the year that an outbreak occurred does not strongly relate in a linear way to the length that it lasted. Therefore, the single year offers limited information about how severe the outbreak was in time.

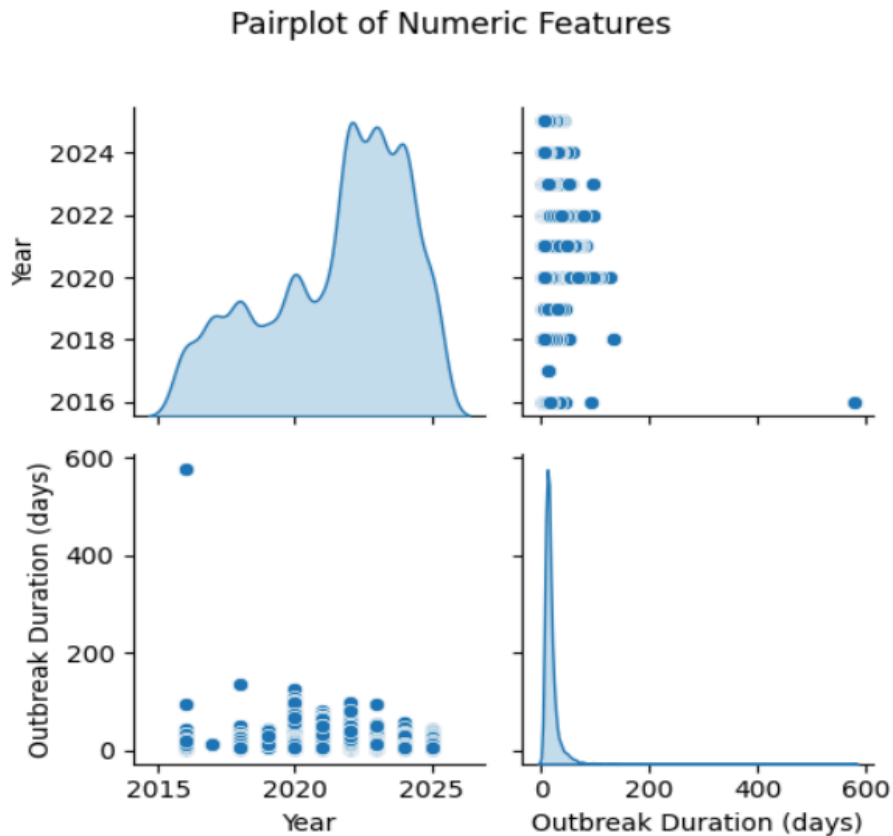


Figure 12 - Pairplot of Numeric Features

The pairplot helped confirm this visually; most outbreaks, no matter the year, tended to stay below 50 days, though the distribution was still skewed with a few long ones. This analysis showed me that outbreak duration is likely influenced more by other features like the type of outbreak, the healthcare setting, or the specific causative agent, rather than by the year alone.

3.3.6 Outlier Detection

Using the interquartile range (IQR) method, I identified 499 outbreaks lasting longer than 32.5 days. Most were respiratory or enteric. I chose to keep them at this stage, as they may represent genuine extended events (e.g., COVID-19) rather than data errors (Figure 13).

```
Lower bound: -3.5 days
Upper bound: 32.5 days
Number of outlier records: 499
```

Sample outliers:

	Year	Outbreak Setting	Type of Outbreak	Outbreak Duration (days)
37	2016	retirement home	enteric	35
69	2016	hospital-chronic care	enteric	34
94	2016	ltch	respiratory	41
127	2016	hospital-acute care	enteric	44
160	2016	ltch	respiratory	34

Figure 13 - Outlier Detection in Outbreak Duration

3.3.7 Group Summaries

```
=== Duration by Type of Outbreak ===
              count      mean  median      std
Type of Outbreak
other              21  29.380952   21.0  21.420729
respiratory       5568  17.315014   14.0  14.267174
enteric           512  13.236328   13.0   8.630761

=== Duration by Outbreak Setting ===
              count      mean  median      std
Outbreak Setting
shelter              3  216.666667   50.0  313.236865
ltch              3391  18.537305   14.0  13.519930
retirement home   1035  16.963285   14.0  10.699003
transitional care    26  15.653846   14.0   7.255025
hospital-acute care  831  14.031288   12.0   9.288235
hospital-chronic care  717  13.237099   12.0   6.813322
hospital-psychiatric  98  12.030612   10.5   5.985298

=== Pivot Table: Avg Duration (days) by Setting x Type ===
Type of Outbreak      enteric      other  respiratory
Outbreak Setting
hospital-acute care   13.014925  29.380952   13.689098
hospital-chronic care  9.517241      NaN   13.393895
hospital-psychiatric  10.125000      NaN   12.200000
ltch                  13.272059      NaN   18.996473
retirement home      14.250000      NaN   17.373749
shelter                NaN      NaN   216.666667
transitional care      NaN      NaN   15.653846
```

Figure 14 - Grouped analysis of Outbreak Duration by Type and Setting

To analyze further, I grouped the data by outbreak type and healthcare setting to see how outbreak durations varied across different categories (Figure 14). When I looked at the type of outbreak, the category labeled as “other” had the highest average duration at around 29 days, followed by respiratory outbreaks at about 17 days, and enteric outbreaks, which tended to be shorter at around 13 days.

Next, I grouped by outbreak setting. While long-term care homes (LTCHs) had the highest number of total outbreaks, the longest durations were actually seen in shelters and retirement homes.

Finally, I used a pivot table to combine both setting and outbreak type. This gave me a clearer picture, for example, respiratory outbreaks in retirement homes and LTCHs were the longest. These group summaries highlighted which combinations of factors lead to prolonged outbreaks.

3.3.8 Temporal & Geospatial Patterns

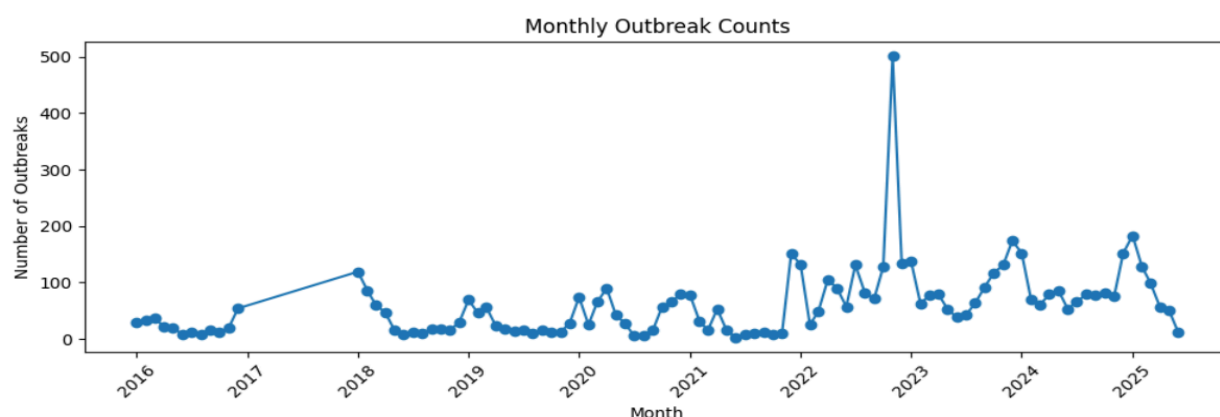


Figure 15 - Monthly Outbreak Counts

This line chart shown in Figure 15 displays monthly outbreak changes between 2016 and 2025. Clear seasonal spikes do exist with the most noticeable surge happening in between late 2022 and early 2023, likely reflecting heightened reporting during COVID-19 waves.

On the geospatial side, I examined which institutions had the most frequent outbreaks. Figure 16 ranks the top ten institutions with the highest outbreak counts. Facilities like Rockcliffe Care Community and Extendicare Scarborough had the highest number of outbreaks, followed by other long-term care and retirement facilities.

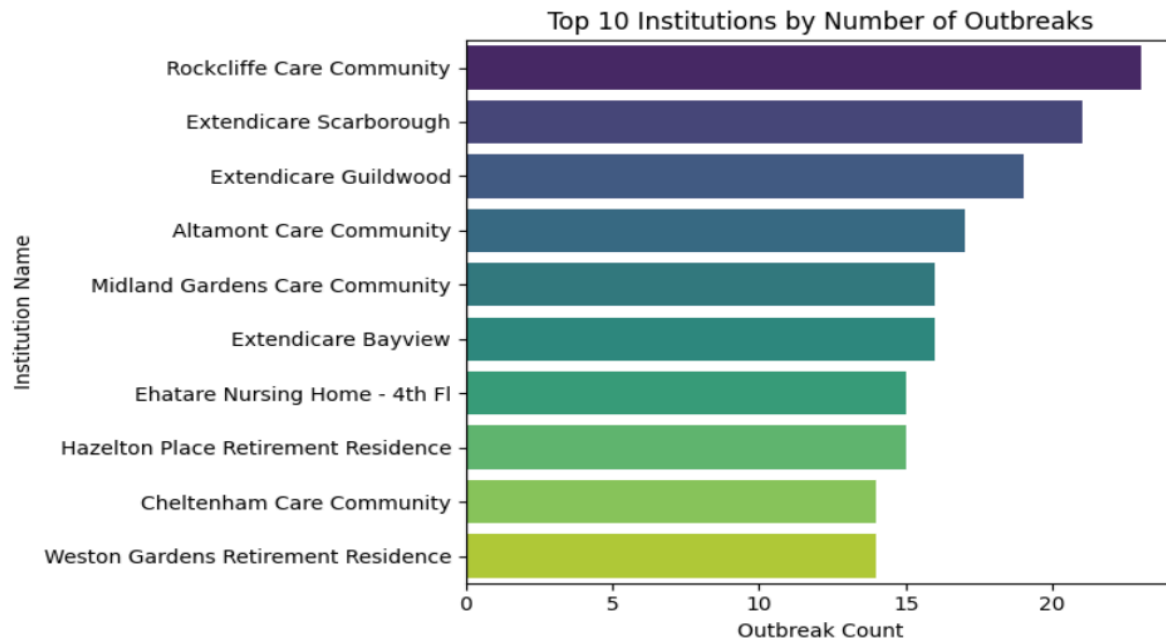


Figure 16 - Top 10 Institutions by Number of Outbreaks

4. FEATURE ENGINEERING

Once the dataset was cleaned up, I extracted out meaningful patterns through feature engineering that could then predict outbreak severity. I created the week's day and month when an outbreak started as temporal features. Seasonal trends could be captured once these were further grouped into seasonal categories such as Winter, Spring, Summer, and Fall. I did use the outbreak duration for categorizing outbreak severity so that outbreaks lasting 7 days or fewer were labeled “Mild,” those between 8 and 21 days were “Moderate,” and those above 21 days were labeled “Severe.” I then encoded these severity labels numerically in order to prepare for modeling.

In addition to that, I introduced binary features to flag whether the outbreak was COVID-related (from either causative agent field) and whether it occurred in a long-term care home. These flags were added because they represent meaningful characteristics that might influence severity. I also applied one-hot encoding to convert categorical variables into numerical format, making them compatible with machine learning algorithms. To ensure that different feature scales wouldn't affect model performance, all numerical features were standardized using z-score normalization. The final dataset, combining the processed features and severity labels, was saved and used for model training.

5. METHODOLOGY AND EXPERIMENTS

5.1 Baseline Model: Logistic Regression

To set a baseline for the evaluating of model performance, a logistic regression model was where I did start. This approach is simple, fast, as well as easy to interpret, so this makes it a useful benchmark. I split the data into training (70%), validation (15%), and test (15%) sets through stratified sampling so that the original distribution of severity classes was maintained. After imputing missing values using the median and scaling all features, I trained the logistic regression model on the training data. While it did not capture complex patterns in the data, it gave me a solid reference point to compare more advanced machine learning and deep learning models.

5.2 Machine Learning Models

When selecting machine learning models, I began by exploring a variety of options commonly used for structured healthcare data. These included Decision Trees, K-Nearest Neighbors, Support Vector Machines, and ensemble methods. After I had considered model complexity along with ease of tuning in addition to ability for handling both categorical and numeric data as well as how well the models generalize within real-world outbreak datasets, I then decided to go forward by using Random Forest and also XGBoost.

Random Forest is a strong ensemble method working through averaging the results of multiple decision trees. This reduces overfitting. It handles class imbalance in a natural way and it shows feature importance. The model was selected since adequate output needs minimal fine-tuning and it's usually dependable.

XGBoost, conversely, is a gradient increasing method that is more powerful. It builds trees in a sequence and it corrects previous mistakes in modeling complex non-linear relationships. I selected it because it performed superiorly in many previous outbreak prediction studies too. I did select it also since we are able to tune it flexibly. I employed randomized cross-validation to fine-tune parameters to optimize model performance. These parameters included learning rate along with maximum tree depth plus the number of estimators.

5.3 Deep Learning Models

To explore more complex non-linear patterns in the data, I also experimented with deep learning models. Initially, I considered options like RNNs, GRUs, and autoencoders. However, I did decide to go along with a Multilayer Perceptron (MLP) plus a 1D Convolutional Neural Network (CNN) after a review of my dataset structure and prior success stories.

The **MLP model, or feedforward neural network**, was selected for its simplicity and strength on tabular data. It consists of fully connected layers and is well-suited for capturing feature interactions that traditional models might miss. This model helped me establish a deep learning baseline and evaluate how well neural networks could perform compared to classical machine learning approaches.

The **CNN model** was included for its ability to capture local feature patterns. Although CNNs are traditionally used in image processing, reshaping structured data into sequences allows CNNs to detect interactions across adjacent features. This property made CNNs a suitable candidate for modeling outbreak severity, as some influencing variables may be related in subtle ways not easily captured by MLPs or tree-based models.

5.4 Hybrid Models

To push beyond individual model capabilities, I explored a hybrid deep learning model that combines CNN and LSTM layers. The CNN component helps extract spatial patterns among input features, while the LSTM component captures temporal dependencies, essentially modeling how patterns evolve over time. The idea was to simulate how outbreaks behave across both space and time by leveraging both types of networks.

While the hybrid model trained successfully and handled frequent severity classes well, its overall performance was somewhat limited. It showed potential, However, the study also made it clear that hybrid models such as CNN+LSTM may require a more refined architecture and better data balancing if they are to truly outperform standalone approaches. Still, the experiment helped to understand hybrid deep learning too. It showed all of the strengths as well as limits for this specific task.

5.5 Ensemble Models

Finally, I then implemented a model ensemble for a combination of CNN and XGBoost prediction probabilities. XGBoost handles data that is structured in a tabular way very well so that fact rationalizes this decision while CNN brings out deeper non-linear interactions. I intended to leverage strengths from both architectures. I chose to mix them rather than depend only on one.

I used a soft voting method here which averaged predicted class probabilities from both models. This ensemble method yielded balanced stable predictions across severity classes, especially when data was imbalanced or slightly noisy. Even though the performance gain wasn't drastic, the consistency in results confirmed that model ensembling can help reduce weaknesses from individual models, as this makes it a promising direction for future outbreak prediction systems.

6. EVALUATION METRICS

For this project, I'm planning to use a combination of evaluation metrics since those metrics go beyond simple accuracy to better reflect the real-world needs of outbreak severity prediction. According to my literature review, many researchers stressed that accuracy alone can be misleading, particularly when the dataset has class imbalance, also this is true in my case as well because "Severe" cases are less frequent.

I intend to focus mainly on the Macro F1-Score, since it equally values each of the class through the averaging of their individual F1-scores. This will help ensure the model performs well along all severity levels and does not favor the most common class.

I intend to assess Precision and Recall per class to better grasp the model's detection of each category. This is particularly critical because to miss a "Severe" case could have serious implications for public health.

I'm planning to use ROC-AUC macro along with AUC-PR macro to assess how well the models separate the severity classes. For probabilistic models, these metrics are helpful to evaluate the confidence as well as discriminative ability particularly in cases of imbalance also widely used throughout recent outbreak prediction studies.

Log Loss is another important metric I plan to include. It is penalizing confident predictions that happen to be incorrect. To know whether my models are well-calibrated and reliable in probability estimates, this will help me to see if they are not just accurate.

Lastly, I'm planning on reporting Cohen's Kappa and Matthews Correlation Coefficient (MCC), as that is my plan. The two metrics better describe model performance when considering agreement beyond chance (Kappa) and when balancing all four quadrants of the confusion matrix (MCC), which helpfully classifies multiple classes.

I plan to assess the models with this full set of metrics, centered on their forecasting power. For models informing public health decisions, evaluating their fairness and confidence is also important.

7. RESULTS

7.1 Logistic Regression (Baseline Model)

```
=== Logistic Regression (Baseline) Validation Metrics ===  
Accuracy: 0.9803  
Macro Precision: 0.9895  
Macro Recall: 0.9508  
Macro F1-score: 0.9690  
AUC-ROC (macro): 0.9990  
AUC-PR (macro): 0.9969  
Log Loss: 0.1128  
Cohen's Kappa: 0.9564  
Matthews CorrCoef: 0.9571
```

Figure 17 - Logistic Regression Validation Metrics

For my baseline, I started with the Logistic Regression model (Figure 17). It gave a solid performance with an accuracy of 98%, which was a good starting point. The macro precision and recall were also high (around 98.9% and 95% respectively), meaning it was generally reliable across all classes. The macro F1-score of 96.9% showed a good balance between precision and recall.

The AUC-ROC and AUC-PR scores were close to perfect, indicating the model was excellent at distinguishing between severity classes. Log loss was low, which means its probability predictions were fairly confident. Cohen's Kappa along with Matthews Correlation Coefficient were also strong because the model's predictions agreed well with the actual outcomes.

In general, Logistic Regression performed in a surprisingly good way. It is a linear model that is simple.

7.2 Random Forest

```
=== Random Forest Validation Metrics ===  
Accuracy: 0.9956  
Macro Precision: 0.9979  
Macro Recall: 0.9892  
Macro F1-score: 0.9935  
AUC-ROC (macro): 1.0000  
AUC-PR (macro): 1.0000  
Log Loss: 0.0895  
Cohen's Kappa: 0.9905  
Matthews CorrCoef: 0.9905
```

Figure 18 - Random Forest Validation Metrics

Next, I trained the Random Forest model (Figure 18), and it clearly outperformed the baseline. It reached an accuracy of about 99.6%, and both macro precision and recall were very high around 99.8% and 98.9% respectively. The F1-score also improved to 99.3%, which shows a strong balance between precision and recall.

Perfect AUC-ROC and AUC-PR scores stood out as major achievements. Due to these scores, it could clearly separate the classes as well as maintain a great balance between sensitivity and specificity. The log loss was even lower than Logistic Regression, which means the model made confident and accurate predictions. Both Cohen's Kappa and MCC were close to 1, showing very strong agreement with actual labels.

Overall, Random Forest gave highly reliable and consistent results across all metrics, with only minor misclassifications in the Severe category.

7.3 XGBoost

```
=== XGBoost Validation Metrics ===  
Accuracy: 1.0000  
Macro Precision: 1.0000  
Macro Recall: 1.0000  
Macro F1-score: 1.0000  
AUC-ROC (macro): 1.0000  
AUC-PR (macro): 1.0000  
Log Loss: 0.0072  
Cohen's Kappa: 1.0000  
Matthews CorrCoef: 1.0000
```

Figure 19 - XGBoost Validation Metrics

The XGBoost model delivered results that were flawless (Figure 19). It achieved 100% accuracy among all severity classes. Perfect scores were obtained then for precision and recall and also for the F1-score. AUC-ROC and AUC-PR both were also perfect since the model was outstanding at separating classes plus handling trade-offs between precision and recall.

The extremely confident predictions by the model made the very low log loss (0.0072) stand out. Cohen's Kappa along with Matthews Correlation Coefficient were both at 1. This balanced performance was in this way then confirmed.

It is clear that by capturing deeper feature patterns, XGBoost outperformed both Random Forest and also Logistic Regression overall. That single model proved most accurate and also reliable within tests.

7.4 Multilayer Perceptron (MLP)

```
=== MLPClassifier Validation Metrics ===  
Accuracy: 0.7158  
Macro Precision: 0.4520  
Macro Recall: 0.5784  
Macro F1-score: 0.4897  
AUC-ROC (macro): 0.8636  
AUC-PR (macro): 0.7266  
Log Loss: 0.8046  
Cohen's Kappa: 0.4205  
Matthews CorrCoef: 0.4427
```

Figure 20 - MLP Classifier Validation Metrics

When I tested the Multilayer Perceptron (MLP), its performance was noticeably lower compared to the tree-based models (Figure 20). The accuracy was around 71.6%, and both macro precision (45.2%) and recall (57.8%) were relatively weak. The F1-score (48.9%) reflected this imbalance too.

The AUC-ROC and AUC-PR values were of a moderate level, but the log loss was of a high one. The model's predictions, therefore, weren't with great confidence. Cohen's Kappa scores and MCC scores noted fair agreement.

Overall, the MLP didn't perform well, especially on the less frequent classes, which suggests that either the architecture or training settings weren't sufficient for this particular dataset.

7.5 1D CNN

```
=== CNN Validation Metrics ===  
Accuracy: 0.7071  
Macro Precision: 0.5555  
Macro Recall: 0.3886  
Macro F1-score: 0.3846  
AUC-ROC (macro): 0.7006  
AUC-PR (macro): 0.4905  
Log Loss: 0.7332  
Cohen's Kappa: 0.1313  
Matthews CorrCoef: 0.1786
```

Figure 21 - CNN Validation Metrics

I also tested a 1D CNN model (Figure 21) to see if it could learn meaningful patterns from the data, but its performance was quite limited. It gave an accuracy of about 70.7%, with macro precision at 55.6% and

recall dropping to just 38.9%. The F1-score stayed low at 38.4%, showing an imbalance between precision and recall.

The AUC-ROC and AUC-PR scores were relatively weak also, and the log loss was high, which suggests that the model wasn't very confident in its predictions. Cohen's Kappa and MCC were quite low so they matched actual labels badly.

The CNN model was worse than even MLP also it did not do well perhaps since data lacked good CNN patterns.

7.6 Hybrid CNN-LSTM

```
=== CNN + LSTM (Hybrid) Validation Metrics ===  
Accuracy: 0.6962  
Macro Precision: 0.3365  
Macro Recall: 0.3370  
Macro F1-score: 0.2873  
AUC-ROC (macro): 0.6923  
AUC-PR (macro): 0.4698  
Log Loss: 0.7439  
Cohen's Kappa: 0.0125  
Matthews CorrCoef: 0.0337
```

Figure 22 - CNN + LSTM (Hybrid) Validation Metrics

For the hybrid model (Figure 22), I combined CNN and LSTM layers to capture both spatial and temporal patterns in the data. However, its performance was the weakest among all models. It gave an accuracy of just 69.6%, and both macro precision and recall were very low (around 33.7%). The F1-score dropped to 28.7%, reflecting the model's struggle across all classes.

AUC-ROC and AUC-PR scores were also limited, and the log loss was quite high, showing the model wasn't confident in its predictions. Both Cohen's Kappa and MCC were near zero, indicating almost no agreement with the true labels.

Overall, the hybrid model didn't generalize well. It likely needed more refined tuning or a larger dataset to fully take advantage of its complex architecture.

7.7 Ensemble Model (XGBoost + CNN)

After evaluating all individual models, I decided to combine the two best-performing ones, XGBoost (from traditional ML) and CNN (from deep learning) to create an ensemble model using soft voting. The

idea was to leverage XGBoost's structured learning strength and CNN's ability to capture deeper patterns, even if its standalone performance was weaker.

```
=== XGB + CNN (Ensemble Val) Validation Metrics ===
Accuracy: 0.9978
Macro Precision: 0.9990
Macro Recall: 0.9946
Macro F1-score: 0.9968
AUC-ROC (macro): 1.0000
AUC-PR (macro): 1.0000
Log Loss: 0.2536
Cohen's Kappa: 0.9952
Matthews CorrCoef: 0.9953

=== XGB + CNN (Ensemble Test) Validation Metrics ===
Accuracy: 0.9967
Macro Precision: 0.9984
Macro Recall: 0.9928
Macro F1-score: 0.9956
AUC-ROC (macro): 1.0000
AUC-PR (macro): 1.0000
Log Loss: 0.2542
Cohen's Kappa: 0.9929
Matthews CorrCoef: 0.9929
```

Figure 23 - XGBoost + CNN (Ensemble) Validation Metrics

The ensemble model (Figure 23) delivered a level of performance that was outstanding. The validation set showed 99.78% accuracy. The precision with recall and also with the F1-score were near perfect too. AUC-ROC as well as AUC-PR both measured a perfect 1.0000 and Cohen's Kappa and MCC scored above 0.99 which indicates extremely high agreement so performance was balanced.

Its performance remained consistently strong on the test set as well, achieving great results across metrics at 99.67% accuracy. Predictions from the ensemble were more stable and generalized. Its log loss was slightly higher than that of XGBoost alone, though.

Overall, this combination proved to be among the most reliable approaches tested in the project, and it confirmed that blending strengths of ML and DL models can lead to strong and high-performing results.

7.8 Comparative Analysis

=== FINAL COMPARISON TABLE ===									
	Logistic Regression (Baseline)	Random Forest	XGBoost	MLPClassifier	CNN	CNN + LSTM (Hybrid)	XGB + CNN (Ensemble Val)	XGB + CNN (Ensemble Test)	
Accuracy	0.9803	0.9956	1.0000	0.7158	0.7071	0.6962	0.9978	0.9967	
Macro Precision	0.9895	0.9979	1.0000	0.4520	0.5555	0.3365	0.9990	0.9984	
Macro Recall	0.9508	0.9892	1.0000	0.5784	0.3886	0.3370	0.9946	0.9928	
Macro F1-score	0.9690	0.9935	1.0000	0.4897	0.3846	0.2873	0.9968	0.9956	
AUC-ROC (macro)	0.9990	1.0000	1.0000	0.8636	0.7006	0.6923	1.0000	1.0000	
AUC-PR (macro)	0.9969	1.0000	1.0000	0.7266	0.4905	0.4698	1.0000	1.0000	
Log Loss	0.1128	0.0895	0.0072	0.8046	0.7332	0.7439	0.2536	0.2542	
Cohen's Kappa	0.9564	0.9905	1.0000	0.4205	0.1313	0.0125	0.9952	0.9929	
Matthews CorCoef	0.9571	0.9905	1.0000	0.4427	0.1786	0.0337	0.9953	0.9929	

Table 1 - Comparative Analysis

The final comparison table (Table 1) highlights clearly the performance differences among all models. Since it was able to manage feature interactions inside structured data XGBoost attained perfect scores across all metrics. In most of the metrics, Random Forest closely trailed behind XGBoost and performed with extremely good results.

The Logistic Regression model performed decently but it had limits modeling relationships that are non-linear. Deep learning models such as MLP, CNN, and CNN+LSTM, consistently did underperform, displaying lower precision, recall, with F1-scores. The dataset's size and lack of sequential patterns have likely contributed to this underperformance.

The ensemble model (XGBoost + CNN) showed metrics nearly identical to XGBoost. It did perform reliably across validation datasets and across test datasets. Tree-based models did generally outperform deep learning models and also reliability and generalization were further improved by the ensemble strategy.

8. CONCLUSION

After testing all the models, I found that XGBoost consistently delivered the best results, because it was in fact accurate and performed in an outstanding way across all metrics. Random Forest performed strongly too. It did prove itself to be just a reliable tree-based method for this structured dataset.

While my Logistic Regression baseline gave a solid starting point, it couldn't capture complex relationships as well as XGBoost. Deep learning models like MLP, CNN, and CNN+LSTM, struggled, likely due to the limited dataset size and lack of strong sequential features.

To balance the strengths of both approaches, I created an ensemble model combining XGBoost and CNN, which produced nearly perfect scores and demonstrated strong generalization on both validation and test sets.

Based on these results, I moved forward with XGBoost and the ensemble model as the core predictive engines. Next I built an interactive dashboard including geospatial and temporal visualizations to present outbreak trends and hotspots.

9. MODEL PREDICTIONS

Using the trained XGBoost model, I generated severity predictions for outbreaks that occurred between May 2025 and May 2026. Table 2 presents a sample of these results, including the institution name, outbreak setting, type of outbreak, causative agents, and the model's predicted severity class. In this example, the model predicted all listed outbreaks as *Mild* (class 0), which aligns with the relatively short durations observed in the corresponding records.

id	Institution Name	Outbreak Setting	Type of Outbreak	Causative Agent-1	Causative Agent-2	Date Outbreak Began	Pred_Class_ID	Pred_Sev
1	The Heritage Nursing Home - 3rd Fl	LTCH	Respiratory	Enterovirus/ Rhinovirus		6/11/2025	0	Mild
2	Cummer Lodge - 2 North	LTCH	Respiratory	Pending		6/10/2025	0	Mild
3	One Kenton Place - 1st & 2nd Fl	Retirement Home	Respiratory	Pending		6/9/2025	0	Mild
4	The O'Neill Centre - 3rd Fl	LTCH	Respiratory	Pending		6/9/2025	0	Mild
5	Rockcliffe Care Community - Ground Fl	LTCH	Respiratory	Respiratory syncytial virus		6/7/2025	0	Mild
6	Chartwell Gibson - 2nd & 4th Fl	LTCH	Respiratory	Rhinovirus		6/6/2025	0	Mild
7	Trillium Health Partners - McCall Centre for Continuing Care - 3rd Fl	Hospital-Chronic Care	Respiratory	COVID-19		6/5/2025	0	Mild
8	Trillium Health Partners - McCall Centre for Continuing Care - 4th Fl	Hospital-Chronic Care	Respiratory	COVID-19		6/3/2025	0	Mild
9	Lakeshore Lodge - 3 South & 3 North	LTCH	Respiratory	Metapneumovirus		6/2/2025	0	Mild
10	Nisbet Lodge - 6th Fl	LTCH	Respiratory	Enterovirus/ Rhinovirus		6/2/2025	0	Mild
11	Belmont House - 2 East	LTCH	Respiratory	Enterovirus/ Rhinovirus		6/1/2025	0	Mild
12	Weston Terrace Care Community - 2A	LTCH	Respiratory	Influenza A (Not subtyped)		5/30/2025	0	Mild
13	Kipling Acres - 1 South Pinepoint	LTCH	Respiratory	Enterovirus/ Rhinovirus		5/27/2025	0	Mild
14	Rockcliffe Care Community - 1st Fl	LTCH	Respiratory	Parainfluenza		5/27/2025	0	Mild
15	Lakeshore Lodge - 2 South	LTCH	Respiratory	Metapneumovirus		5/26/2025	0	Mild
16	Baycrest Apotex Centre - Jewish Home for the Aged - 2E1	LTCH	Respiratory	Metapneumovirus		5/26/2025	0	Mild
17	Wesburn Manor - 4 North	LTCH	Respiratory	Rhinovirus		5/26/2025	0	Mild
18	Kensington Gardens (South Building) - 1W, 1E & 4E	LTCH	Respiratory	Coronavirus*		5/25/2025	0	Mild
19	Kennedy Lodge - 1st & 2nd Fl	LTCH	Respiratory	Parainfluenza		5/24/2025	0	Mild

Table 2 - Predictions with Xgboost model from May 2025 - May 2026

After developing the XGBoost and CNN models, I combined them using a soft-voting ensemble approach to leverage the strengths of both architectures.

id	Institution Name	Outbreak Setting	Type of Outbreak	Causative Agent-1	Causative Agent-2	Date Outbreak Began	Pred_Class_ID	Pred_Sev
1	The Heritage Nursing Home - 3rd Fl	LTCH	Respiratory	Enterovirus/ Rhinovirus		6/11/2025	0	Mild
2	Cummer Lodge - 2 North	LTCH	Respiratory	Pending		6/10/2025	0	Mild
3	One Kenton Place - 1st & 2nd Fl	Retirement Home	Respiratory	Pending		6/9/2025	0	Mild
4	The O'Neill Centre - 3rd Fl	LTCH	Respiratory	Pending		6/9/2025	0	Mild
5	Rockcliffe Care Community - Ground Fl	LTCH	Respiratory	Respiratory syncytial virus		6/7/2025	0	Mild
6	Chartwell Gibson - 2nd & 4th Fl	LTCH	Respiratory	Rhinovirus		6/6/2025	0	Mild
7	Trillium Health Partners - McCall Centre for Continuing Care - 3rd Fl	Hospital-Chronic Care	Respiratory	COVID-19		6/5/2025	0	Mild
8	Trillium Health Partners - McCall Centre for Continuing Care - 4th Fl	Hospital-Chronic Care	Respiratory	COVID-19		6/3/2025	0	Mild
9	Lakeshore Lodge - 3 South & 3 North	LTCH	Respiratory	Metapneumovirus		6/2/2025	0	Mild
10	Nisbet Lodge - 6th Fl	LTCH	Respiratory	Enterovirus/ Rhinovirus		6/2/2025	0	Mild
11	Belmont House - 2 East	LTCH	Respiratory	Enterovirus/ Rhinovirus		6/1/2025	0	Mild
12	Weston Terrace Care Community - 2A	LTCH	Respiratory	Influenza A (Not subtyped)		5/30/2025	0	Mild
13	Kipling Acres - 1 South Pinepoint	LTCH	Respiratory	Enterovirus/ Rhinovirus		5/27/2025	0	Mild
14	Rockcliffe Care Community - 1st Fl	LTCH	Respiratory	Parainfluenza		5/27/2025	0	Mild
15	Lakeshore Lodge - 2 South	LTCH	Respiratory	Metapneumovirus		5/26/2025	0	Mild
16	Baycrest Apotex Centre - Jewish Home for the Aged - 2E1	LTCH	Respiratory	Metapneumovirus		5/26/2025	0	Mild
17	Wesburn Manor - 4 North	LTCH	Respiratory	Rhinovirus		5/26/2025	0	Mild
18	Kensington Gardens (South Building) - 1W, 1E & 4E	LTCH	Respiratory	Coronavirus*		5/25/2025	0	Mild
19	Kennedy Lodge - 1st & 2nd Fl	LTCH	Respiratory	Parainfluenza		5/24/2025	0	Mild

Table 3 - Predictions with Ensemble model from May 2025 - May 2026

Table 3 shows a sample of the ensemble model's predictions for outbreaks from May 2025 to May 2026. As with the XGBoost results, the ensemble model predicted these outbreaks as Mild (class 0), consistent with their shorter durations and less severe patterns in the data.

10. LIMITATIONS AND FUTURE WORK

While this project gave me some promising results in predicting outbreak severity, there are a few limitations I need to acknowledge. The dataset covered multiple years yet had far fewer severe cases in comparison to mild and moderate ones. Rare yet key events patterns may have been harder for the models to learn. Important external factors such as weather and population density as well as mobility patterns were missing, so the features that I used were mostly outbreak-related metadata, which could then have added valuable context. Because of novel pathogens or shifting health habits, the models may also battle adapting if future outbreaks differ greatly; they were trained using past data.

Tree-based models such as XGBoost did work better than deep learning models on the modeling side for the reason that the dataset was not large enough for neural networks in order to shine. Transparency was also absent in certain ensemble deep learning components. Public health teams might find trusting them harder now.

In the future, I'd like to expand the dataset with additional sources such as environmental as well as demographic data, and I want to build a real-time updating pipeline so predictions stay relevant. Explainability tools like SHAP should be added to make predictions more transparent in addition to exploring class imbalance methods and advanced feature engineering. In outbreak management, I can see this evolving with time into a more strong, explainable, real-time decision-support tool.

11. APPENDIX

All source code and evaluation scripts are made publicly available for transparency and reproducibility.

The complete implementation can be accessed at:

https://github.com/RitikaAggarwal22/MRP_Project

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